LieRing Programmer Manual

modules, scripts, accepted design choices

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1 Programmer Manual

Implementation notes for the LieRing Julia package. Physics is in the Beam Physics Manual.

1.1 Layout

path	role
`src/Constants.jl`	CODATA deuteron, reference particle, $v(\delta)$, $e(\delta)$, exact $z \leftrightarrow u$
`src/Jets.jl`	TaylorN space, `JET_ORDER=4`, `MAP_ORDER=3`
`src/Elements.jl`	all element types + lattice-editing helpers
`src/Hamiltonians.jl`	element H as polynomials
`src/LieOperators.jl`	Poisson bracket, exp(: f :), Dragt–Finn
`src/OptiMParser.jl`	`.opt` reader (mass from particle, not file header)
`src/LinearOptics.jl`	Twiss, dispersion, tunes, α_0
`src/Symplectic.jl`	$J^\mathsf{T} S J - S$
`src/Tracking.jl`	factorised Lie tracking, thin RF kick
`src/TaylorMap.jl`	DA (RK4) one-turn map
`src/NormalForm.jl`	linear normalisation (nonlinear: planned)
`src/SecondOrder.jl`	D_2, $\alpha_{0,1}$, `blond_alphas`, BLonD drifts, separatrix
`src/ExactIntegrator.jl`	exact-H integrator, finite cavity, renormalise
`src/Acceleration.jl`	ramp, tune fit, turn snapshots
`src/Plotting.jl`	figures and GIFs
`src/Optimizers.jl`	optional sextupole response / example solves

Entry: `include("src/LieRing.jl"); using .LieRing`.

1.2 Scripts

```
julia --project=. scripts/01_linear_optics.jl
julia --project=. scripts/02_lie_maps.jl
julia --project=. scripts/03_second_order.jl
julia --project=. scripts/04_optional_sextupole_optimization.jl
julia --project=. scripts/05_rf_acceleration.jl
julia --project=. scripts/06_coordinates.jl
julia --project=. scripts/07_blond_longitudinal.jl
```

1.3 Accepted design choices

- Injection kinetic energy 10 MeV; final 270 MeV from the lattice header. - RF: $L = 1$ m, $V = 0.2$ MV, $h = 5$, $\phi_s = 150^\circ$ (ramp) / 180° (flat). - Mass = CODATA deuteron 1875.61294257 MeV (OptiM `Mass=2*938.272` ignored). - Lattice sequence used once (file `Number of periods=2` is an error; bend is 360 deg). - Closed $D(0) = 0.011$ m (OptiM initial 0 is not the closed solution). - Test amplitudes: $x_0 = y_0 = 2$ mm, $\delta_0 = 10^{-3}$; ensemble as in script 05. - Optimiser targets A/B are examples only; default sextupole strengths are zero.

1.4 Docs build

```
julia --project=docs docs/make.jl          # HTML -> docs/build/
julia docs/make_pdf.jl                     # Beam Physics PDF
julia docs/make_pdf_programmer.jl          # Programmer PDF
```